使用公开聚合的语义图数据和图神经网络改进预测毒理学的QSAR建模

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定量构效关系（QSAR）建模是预测化学毒性最常用的计算技术，但QSAR方法缺乏创新导致其性能不尽如人意。我们展示了将来自开放获取公共数据库的语义图数据整合到现代QSAR建模中，并在图神经网络（GNNs）的框架下分析这些数据，可以显著提升预测毒理学的效果。此外，我们对GNNs进行了内部分析，以展示它们如何使QSAR的应用更具可解释性，并通过消融分析探讨不同数据元素对最终模型性能的贡献。

关键词：毒理学；图神经网络；数据聚合；QSAR；人工智能。

1. 介绍

评估化学品的毒性是药物和环境研究的重要组成部分。传统上，确定毒性的任务是通过体内模型完成的，即让模型生物暴露于感兴趣的化学物质中并观察其毒性效应，或者通过对人群进行流行病学研究。以上两种方法既昂贵又耗时，鉴于数十万种具有毒理学研究价值的化合物，需要创新的方法来快速筛选化学品。近年来，预测性毒理学和大规模化学筛选工作已应运而生，以应对这一问题。

定量构效关系（QSAR）建模可以说是预测化学物质是否会在体外引起毒性反应的最常用方法。简而言之，QSAR 建模包括收集分子结构的规则量化描述（称为指纹），然后将统计模型（例如逻辑回归、随机森林等）拟合到已知感兴趣毒性终点的化学物质集合上。由于用于训练模型的每个数据点本身都是单个实验的结果，QSAR 是一种元分析方法，其复杂性不仅在于捕捉化学物质的相关结构特征的挑战，还在于用于生成训练数据的基础实验中的错误、偏差和歧义。因此，QSAR 经常因其在许多任务上的表现令人失望而受到批评。计算毒

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Improving QSAR Modeling for Predictive Toxicology using Publicly Aggregated Semantic Graph Data and Graph Neural Networks

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Quantitative Structure-Activity Relationship (QSAR) modeling is the most common computational technique for predicting chemical toxicity, but a lack of methodological innovations in QSAR have led to underwhelming performance. We show that contemporary QSAR modeling for predictive toxicology can be substantially improved by incorporating semantic graph data aggregated from open-access public databases, and analyzing those data in the context of graph neural networks (GNNs). Furthermore, we introspect the GNNs to demonstrate how they can lead to more interpretable applications of QSAR, and use ablation analysis to explore the contribution of different data elements to the final models' performance.

Keywords: Toxicology; Graph neural networks; Data aggregation; QSAR; Artificial intelligence.

1. Introduction

Evaluating the toxicity of chemicals is an essential component of pharmaceutical and environmental research. Traditionally, the task of establishing toxicity has been accomplished using *in vivo* models, where a model organism is exposed to a chemical of interest and observed for toxic effects, or by performing epidemiological studies on human populations. Both of these approaches are costly and time consuming,[1] and given the hundreds of thousands of compounds of toxicological interest, innovative alternatives are needed to rapidly screen chemicals. In recent decades, predictive toxicology and large-scale chemical screening efforts have emerged to address this issue.[2,3]

Quantitative Structure-Activity Relationship (QSAR) modeling is arguably the most prevalent method for predicting *in silico* whether a chemical will cause a toxic response.[4] Briefly, QSAR modeling involves collecting regularly structured quantitative descriptions of molecular structures (known as *fingerprints*), and then fitting a statistical model (e.g., logistic regression, random forest, etc.) to sets of chemicals where a toxic endpoint of interest is already known.[5,6] Since each data point used to train a model is itself the outcome of a single experiment, QSAR is a meta-analysis approach that is complicated not only by the challenge of capturing relevant structural features of chemicals, but also by errors, biases, and ambiguities in the underlying experiments used to generate the training data. Consequently, QSAR is often criticized for its disappointing performance on many tasks.[7,8] The computational tox-

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毒理学界长期以来一直承认需要新的方法创新来提高QSAR性能，但很少有人能够有效实施。

在本研究中，我们通过结合来自多个公共数据源的多模态图数据来增强传统的QSAR方法，并在异构图卷积神经网络（GCNN）模型的背景下分析这些数据，从而解决这些问题。我们使用Tox21数据存储库中的52个测定及其伴随的化学筛选数据来评估模型，并将其性能与由随机森林和梯度提升分类器组成的两个严格定义的传统QSAR模型进行比较。我们的结果显示，GNN策略显著优于传统QSAR。我们进一步通过移除图的各个组件来优化结果，以解释不同数据源对GNN性能提升的相对贡献。最后，我们讨论了GNN如何提高QSAR的可解释性，并提出了继续这一研究工作的未来方向。

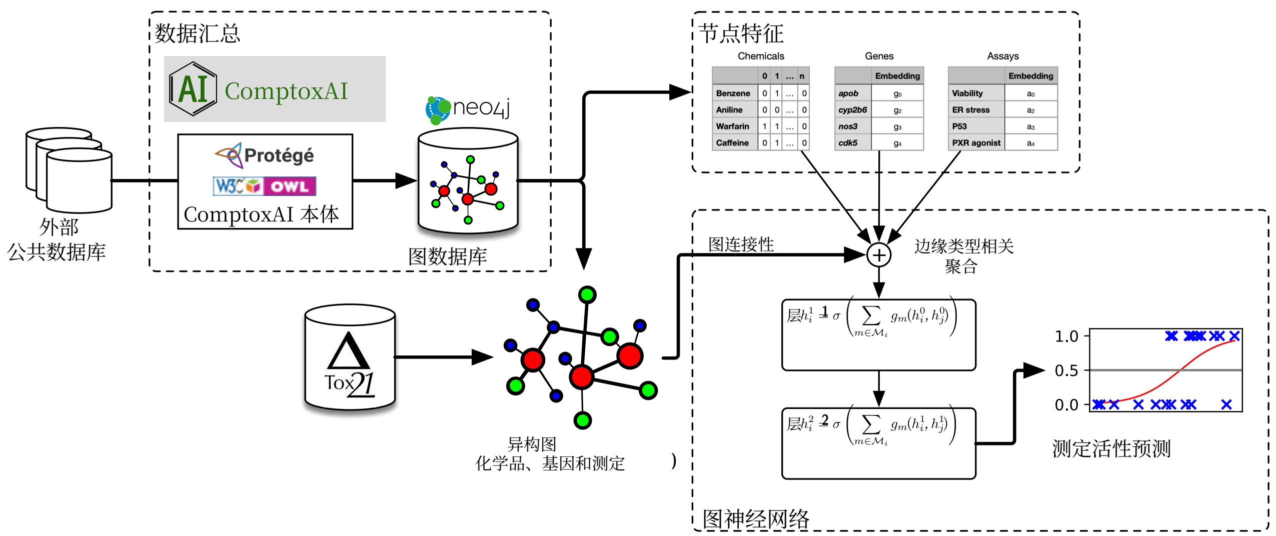

图1. 本研究中使用的图机器学习方法概述。我们构建了一个以毒理学为核心的图数据库（命名为ComptoxAI），使用从各种公共数据库汇总的数据，并提取用于QSAR分析的子图，包含化学物质、实验和基因。然后，我们训练并评估一个图神经网络，以预测某种化学物质是否会激活来自Tox21数据库的特定毒理学实验。

2. 方法

2.1. 获取毒理学检测数据

我们使用了 Tox21 数据集[2]——这是由美国国立卫生研究院、美国食品药品监督管理局和美国环境保护局合作生成的免费资源——以获取用于分类的一组候选检测方法，并建立特定化学品与这些检测方法之间的“真实关系”。数据库中的每个检测方法都包括描述该检测方法活性的实验筛选结果

icology community has long acknowledged the need for new methodological innovations to improve QSAR performance, but few have been effectively implemented.

In this study, we address these issues by augmenting the traditional QSAR approach with multimodal graph data aggregated from several public data sources, and analyzing those data in the context of a heterogeneous graph convolutional neural network (GCNN) model. We evaluate the model using 52 assays and their accompanying chemical screening data from the Tox21 data repository, and compare its performance to two rigorously defined traditional QSAR models consisting of random forest and gradient boosting classifiers. Our results show that the GNN strategy significantly outperforms traditional QSAR. We further refine our results by removing various components of the graph to explain the relative contributions of different data sources to the GNNs' increased performance. Finally, we discuss how GNNs improve the interpretability of QSAR, and suggest future directions to continue this body of work.

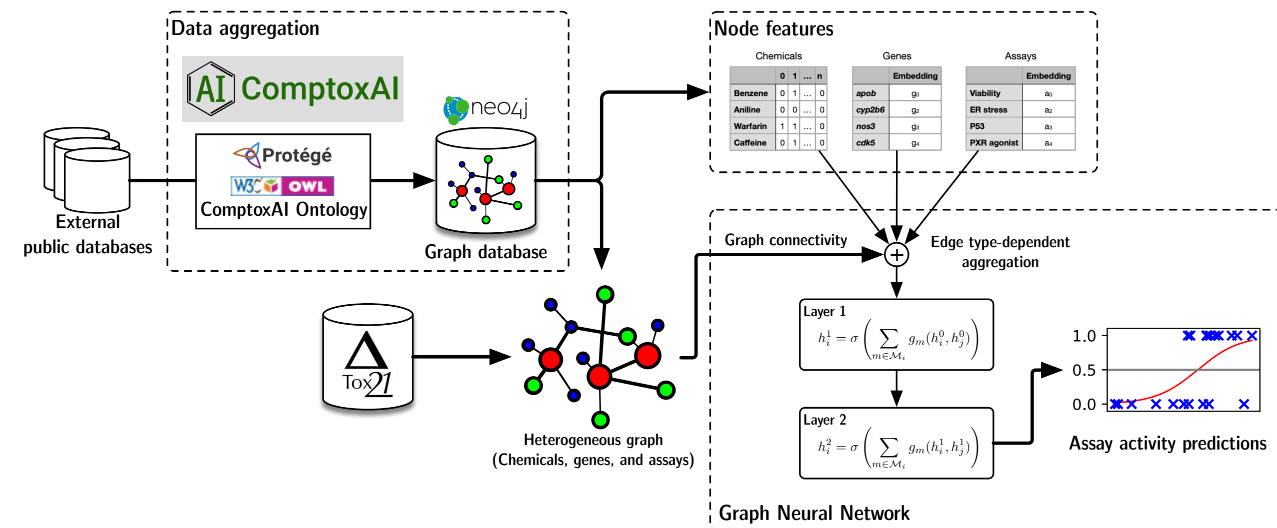

Fig. 1. Overview of the graph machine learning approach used in this study. We build a toxicology-focused graph database (named ComptoxAI) using data aggregated from diverse public databases, and extract a subgraph for QSAR analysis containing chemicals, assays, and genes. We then train and evaluate a graph neural network that predicts whether or not a chemical activates specific toxicology-focused assays from the Tox21 database.

2. Methods

2.1. *Obtaining toxicology assay data*

We used the Tox21 dataset[2]—which is a freely available resource produced collaboratively by the US National Institutes of Health, the US Food and Drug Administration, and the US Environmental Protection Agency—to obtain a set of candidate assays for classification and establish 'ground truth' relationships between specific chemicals and those assays. Each assay in the database includes experimental screening results describing the activity of the assay in

对具有毒理学关注的特定化学物质的反应，包括药物、小分子代谢物、环境毒素等。我们删除了所有结果不确定或模糊的化学-检测测量，以及活性化学物质非常少（例如，< 100）的检测。

2.2. 聚合公开可用的多模态图数据

本研究中使用的图数据来自一个新的计算毒理学数据资源，名为 ComptoxAI[a]。ComptoxAI 包含一个大型图数据库，其中包含许多与毒性转化机制相关的实体和关系类型，所有数据均来源于第三方公共数据库（包括 PubChem、Drugbank、美国环保署的计算毒理学仪表板、NCBI 基因等）。我们从 ComptoxAI 的图数据库中提取了子图，该子图定义为所有表示化学物质、基因和毒理学测定的节点，以及连接这些类型节点的完整边集合。

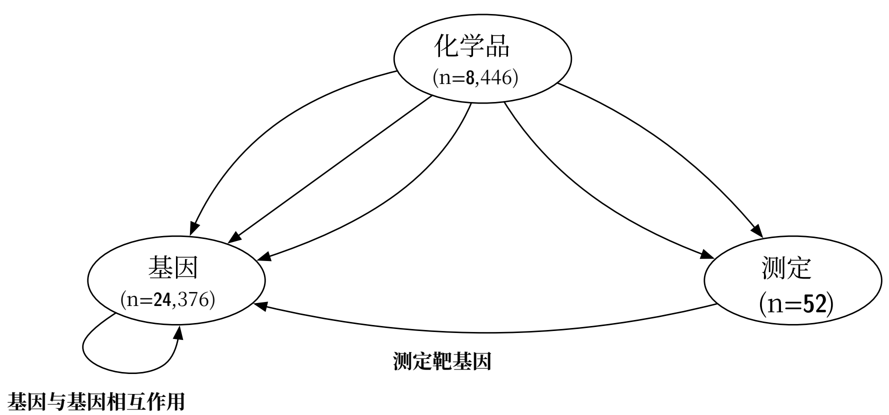

图 2. 元图描述了异构图中的节点类型、节点数量和边类型。在 GNN 的实现过程中，我们还定义了相应的反向边（例如，assay-TargetsGene ↔ geneTargetedByAssay），以便促进 GNN 的消息传递范式。

提取的子图节点由三种实体类型组成：化学物质、测定和基因。我们从美国环保局的DSSTox数据库中获取化学物质，[9] 并对其进行了进一步筛选，以确保每种化学物质都对应于PubChem中的一个独特化合物。我们从NCBI基因数据库获取基因，[10] 并如上所述从Tox21筛选库获取测定。为了作为化学物质的节点特征，我们使用其SMILES字符串计算了图中所有化学物质的MACCS化学描述符指纹，[11] 。每个指纹由长度为166的位串组成，每个位表示特定化学特征的存在或不存在。这些指纹也被重复用作下文描述的基线（非GNN）QSAR模型的预测特征。图中的所有边都来源于Hetionet数据库[12] 或Tox21中的测定–化学注释。子图中节点和边类型的元图如图2所示。

[a]ComptoxAI 的完整图数据库可以在 https://comptox.ai 找到，并将在即将发布的另一篇文章中进行描述。

response to specific chemicals of toxicological interest, including pharmaceutical drugs, small molecule metabolites, environmental toxicants, and others. We removed all chemical–assay measurements with inconclusive or ambiguous results, as well as assays with very few (e.g., < 100) active chemicals.

2.2. *Aggregating publicly available multimodal graph data*

The graph data used in this study come from a new data resource for computational toxicology, named ComptoxAI[a]. ComptoxAI includes a large graph database containing many entity and relationship types that pertain to translational mechanisms of toxicity, all of which are sourced from third-party public databases (including PubChem, Drugbank, the US EPA's Computational Toxicology Dashboard, NCBI Gene, and many others). We extracted the subgraph from ComptoxAI's graph database defined as all nodes representing chemicals, genes, and toxicological assays, as well as the complete set of edges linking nodes of those types.

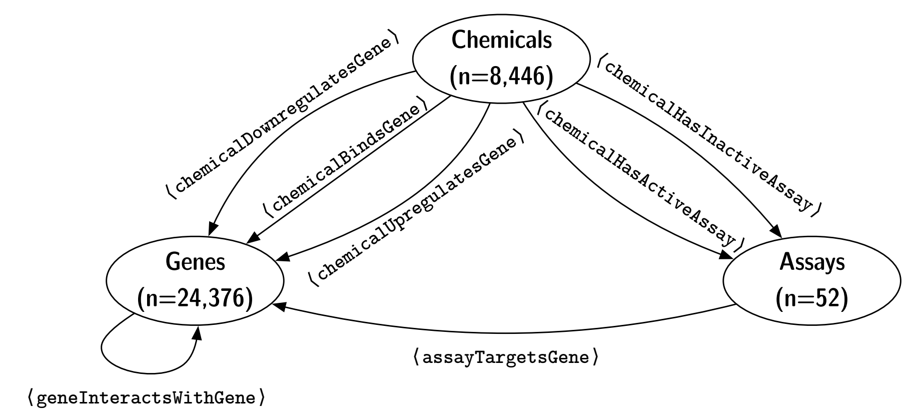

Fig. 2. Metagraph describing the node types, node counts, and edge types in the heterogeneous graph. During implementation of the GNN, we also define corresponding inverse edges (e.g., assay-TargetsGene ↔ geneTargetedByAssay) to facilitate the message-passing paradigm of the GNN.

The 3 entity types that comprise the nodes of the extracted subgraph are *chemicals*, *assays*, and *genes*. We sourced the chemicals from the US EPA's DSSTox database,[9] and further filtered them so that each one is equivalent to a distinct compound in PubChem. We obtained genes from the NCBI Gene database,[10] and assays from the Tox21 screening repository as described above. To serve as node features for chemicals, we computed MACCS chemical descriptor fingerprints[11] for all chemicals in the graph, using their SMILES strings. Each fingerprint is comprised of a bit-string of length 166, where each bit indicates presence or absence of a specific chemical characteristic. These fingerprints are also reused as predictive features in the baseline (non-GNN) QSAR models, described below. All edges in the graph were sourced from either the Hetionet database[12] or from assay–chemical annotations in Tox21. A metagraph describing the node and edge types in the subgraph is shown in **Figure 2**.

[a]The full graph database for ComptoxAI can be found at `https://comptox.ai`, and will be described in a separate, upcoming publication.

2.3. 异构图神经网络

我们为图机器学习实验构建了一个异构图卷积神经网络（GCNN）架构。由于我们的图包含多种实体类型（化学品、基因和测定）——每种类型可能具有不同的节点特征集合，并通过多个语义不同的边类型相连接——该架构扩展了常见的GCNN模型，以为每种边类型学习不同的信息传递函数。简而言之，网络的每一层都会聚合来自图中相邻节点的信号，因此更多的层意味着信号将从每个节点周围更广的范围聚合。网络的输出可以被视为节点的编码表示，包含其局部邻域中其他节点的信息。GCNN也可以被看作是卷积神经网络（CNN）的一种推广——与在计算机视觉中使用的卷积算子从图像中邻近像素聚合信号不同，它聚合

在异构图中，不同的节点类型代表不同类型的实体，每种节点都在语义上不同的特征空间中表示。因此，从相邻节点聚合信息的过程必须考虑这些节点的类型。此外，不同的边类型（例如，chemicalUpregulatesGene和chemicalDownregulatesGene）传递它们各自语义上不同的含义，这可能会显著影响信息在网络中的传递。为了解决这两个挑战，我们为图中每种边类型学习单独的聚合函数，遵循Schlichtkrull等人在R-GCN（关系图卷积网络）中提出的示例。在R-GCN范式下，消息传递过程可以分为三个连续的步骤：(1.) 使用适当的边类型特定消息函数收集来自相邻节点的信号，(2.) 通过一个聚合函数将所有这些传入信号（跨所有边类型）进行组合

GNN 的正式描述见附录 A。

2.3.1. 节点分类

根据上文描述的GCNN架构，我们构建了一个异质图，其中化学物质根据它们是否（1）或是否不（0）激活感兴趣的检测方法来标记。虽然我们移除了代表感兴趣检测方法的节点[b]，但所有其他Tox21检测方法都包含在图中，因此化学物质与其他检测方法之间的边可以用来提高模型的推断能力，超过基线QSAR模型的能力，后者只能访问化学结构。同样，与基因相关的相互作用进一步增加了模型可用的信息。我们使用MACCS指纹作为节点特征，而检测方法和基因节点初始化为1维均匀随机值，这些值在模型训练过程中被优化，最终作为标量“嵌入”，大致与这些节点在训练网络中的重要性成比例。该过程

[b]为了防止信息泄露，因为与检测的连通性会完美预测节点标签。

2.3. *Heterogeneous graph neural network*

We constructed a heterogeneous graph convolutional neural network (GCNN) architecture[13] for the graph ML experiments. Since our graph contains multiple entity types (chemicals, genes, and assays)—each with possibly different sets of node features, and linked by multiple semantically distinct edge types—the architecture extends the common GCNN model to learn separate message passing functions for each edge type. Briefly, each layer of the network aggregates signals from adjacent nodes in the graph, such that a greater number of layers results in signals being aggregated from an increasingly wider radius around each node. The output of the network can be thought of as encoded representations of nodes that incorporate information from the other nodes in their local neighborhood. The GCNN can also be thought of as a generalization of convolutional neural networks (CNNs) used in computer vision—instead of the convolutional operator aggregating signals from nearby pixels in an image, it aggregates features from adjacent nodes in the graph.[14]

In a heterogeneous graph, different node types represent different types of entities, each represented within a semantically distinct feature space.[15] Therefore, the process of aggregating information from adjacent nodes must take those nodes' types into account. Additionally, different edge types (e.g., ⟨`chemicalUpregulatesGene`⟩ and ⟨`chemicalDownregulatesGene`⟩) convey their own semantically distinct meanings, which can substantially effect the flow of information through the network. To handle these two challenges, we learn separate aggregation functions for each edge type in the graph, following the example proposed by Schlichtkrull *et al* in R-GCNs (Relational Graph Convolutional Networks).[16] Within the R-GCN paradigm, the message passing process can be split into 3 sequential steps: (1.) collecting signals from adjacent nodes using an appropriate edge type-specific message function ϕ, (2.) combining each of those incoming signals (across all edge types) via a reduce function ρ, and (3.) finally updating the target node v by applying an update function ψ. Training the network is roughly equivalent to finding an appropriate parameterization of ϕ for each edge type.

A formal description of the GNN is given in **Appendix A**.

2.3.1. *Node classification*

Given the GCNN architecture described above, we construct a heterogeneous graph where chemicals are labeled according to whether they do (1) or do not (0) activate an assay of interest. Although we remove the node representing the assay of interest[b], all other Tox21 assays are included in the graph, and edges between chemicals and the other assays can therefore be used to improve the inferential capacity of the model beyond those of the baseline QSAR models, which only have access to chemical structure. Similarly, interactions involving genes further increase the information available to the model. We use the MACCS fingerprints as node features, while assay and gene nodes are initialized as 1-dimensional uniform random values that are optimized during model training, eventually serving as scalar 'embeddings' that are roughly proportional to those nodes' importance in the trained network. The procedure

[b]To prevent information leakage, since conectivity to the assay would perfectly predict the node labels.

用于标注图的我们在算法1中展示。

算法1 有标记的异构图构建用于毒性测定QSAR模型。

设 G 为 QSAR 的异质图， $a \in \mathcal{A}$ 为感兴趣的测定， $\ell(c)$ 表示关于测定 a 的化学物质 $c \in \mathcal{C}$ 的活性标签。
对于每种化学物质 $c \in \mathcal{C}$ 执行
　如果 $\exists$ **有一条边** (c, r, a)**，且** r **的边类型为** $\langle$**chemicalHasActiveAssay**$\rangle$**，则**
　　$\ell(c) \leftarrow 1$
　否则如果 $\exists$ **存在一条边** (c, r, a)**，使得** r **的边类型是** $\langle$ **chemicalHasInactiveAssay** $\rangle$ **然后**
　　$\ell(c) \leftarrow 0$
　else
　　$\ell(c) \leftarrow$ 未定义 ▷ 节点 c 没有可用标签
　结束如果
结束
$G_a^\star \leftarrow \mathcal{G} \setminus a$ 从图中删除节点 a 以防止信息泄露
返回 $G^\star{}_a$

然后，将包含标记化学物质的生成图 $G_a^\star$ 用作 GCNN 的输入，我们训练它以预测正确的标签。我们对标记化学物质使用 80%/20% 的训练/测试划分，使用 Adam 算法（适用于稀疏梯度的计算效率高的随机梯度下降变体）优化 GCNN 的参数 [17] ，并通过交叉熵损失计算预测标签与真实标签之间的误差。

附加 节点分类方法的详细信息见附录B .

2.4. 基线*QSAR*分类器

为了评估GNN分类模型的相对性能，我们构建了两个额外的（非神经网络）QSAR模型，这些模型代表了与当前预测毒理学实践一致的严格定义基准：一个随机森林分类器， [18] 和一个梯度提升分类器。[19] 每个模型都使用上述从SMILES字符串计算的化学品MACCS指纹进行训练，训练/测试集比例为80%/20%。我们为每个随机森林模型调节了6个超参数，为每个梯度提升模型调节了5个超参数，如表S1所述。超参数通过网格搜索进行调节，最优超参数集定义为在训练数据上使预测标签与真实标签之间的二元交叉熵最小化的参数组合。

3. 结果

3.1. G 神经网络节点分类性能与基线*QSAR*模型比较 *els*

在Tox21数据库中的68个检测中，我们保留了52个用于QSAR实验的分析。其余16个检测没有被使用，这是因为活性化学物质数量较少或在ComptoxAI图数据库中筛选的化学物质代表性不足。此外，我们丢弃了筛选结果不明确或模糊的化学物质的化合物标签。

we use for labeling the graph is shown in **Algorithm 1**.

Algorithm 1 Labeled heterogeneous graph construction for toxicity assay QSAR model.

Let G be a heterogeneous graph for QSAR, $a \in \mathcal{A}$ be an assay of interest, and $\ell(c)$ denote an activity label for chemical $c \in \mathcal{C}$ w.r.t. assay a.
for each chemical $c \in \mathcal{C}$ **do**
　if $\exists$ an edge (c, r, a) s.t. the edge type of r is $\langle$`chemicalHasActiveAssay`$\rangle$ **then**
　　$\ell(c) \leftarrow 1$
　else if $\exists$ an edge (c, r, a) s.t. the edge type of r is $\langle$`chemicalHasInactiveAssay`$\rangle$ **then**
　　$\ell(c) \leftarrow 0$
　else
　　$\ell(c) \leftarrow$ is undefined ▷ No label available for node c
　end if
end for
$G_a^\star \leftarrow \mathcal{G} \setminus a$ ▷ Delete node a from the graph to prevent information leakage
return $G_a^\star$

The resulting graph $G_a^\star$ containing labeled chemicals is then used as input to the GCNN, which we train to predict the correct labels. We use an 80%/20% train/test split on the labeled chemicals, optimize the GCNN's parameters using the Adam algorithm (a computationally efficient variant of stochastic gradient descent suitable for sparse gradients),[17] and compute the error between predicted and true labels via cross entropy loss.

Additional details on the node classification approach are given in **Appendix B**.

2.4. *Baseline QSAR classifiers*

To assess the relative performance of the GNN classification model, we built 2 additional (non-NN) QSAR models that represent rigorously defined benchmarks consistent with current practice in predictive toxicology: A random forest classifier,[18] and a gradient boosting classifier.[19] Each model was trained on the aforementioned MACCS fingerprints of chemicals computed from SMILES strings, with an 80%/20% training/testing split. We tuned 6 hyperparameters for each random forest model, and 5 for each gradient boosting model, as described in **Table S1**. These were tuned using grid search, where the optimal hyperparameter set is defined as the one that minimized binary cross entropy between predicted labels and true labels on the training data.

3. Results

3.1. *GNN node classification performance vs. baseline QSAR models*

Of the 68 total assays in the Tox21 database, we retained 52 for analysis in the QSAR experiments. The remaining 16 assays were not used due to either a low number of active chemicals or underrepresentation of screened chemicals in the ComptoxAI graph database. Additionally, we discarded compound labels for chemicals with inconclusive or ambiguous screening results.

如图3所示，GNN模型在受试者工作特征曲线下面积（AUROC）方面显著优于随机森林（Wilcoxon符号秩检验 p-值 $2.3 \cdot 10^{-4}$）和梯度提升（p-值 $2.6 \cdot 10^{-3}$）模型，平均AUROC为0.883（随机森林为0.834，梯度提升为0.851）。这是一个有力的证据，表明GNN模型往往显著优于"传统"QSAR模型。GNN AUROC的一个显著特征是其分布的方差高于随机森林或梯度提升的AUROC。根据经验，这可能是因为在正样本较少的实验上训练时，GNN模型的敏感性下降——神经网络在数据稀疏时往往表现不佳，似乎正是这种情况。我们还比较了三种模型类型的F1-score分布；然而，三种模型之间的差异在统计学上并不显著。相对较低的F1-scores

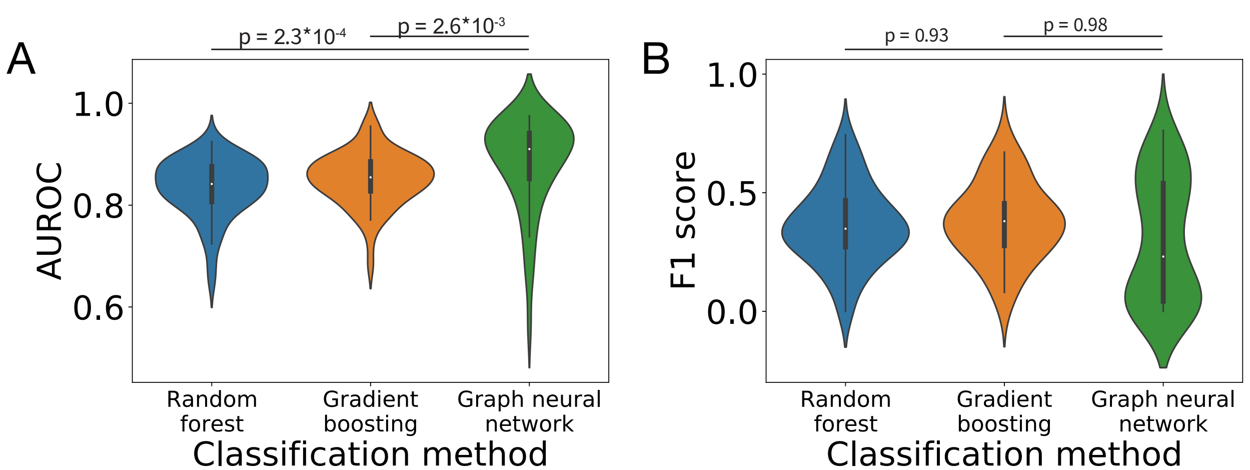

图3. 三种QSAR模型类型在每个Tox21测定上的整体性能指标——a.) 受试者工作特征曲线下面积（AUROC）和b.) F1分数。GNN模型的平均AUROC显著高于任何一种基线QSAR方法。F1分数的差异在统计上不显著。GNN在Tox21中注释"活性"较少（例如， < 100）的测定上F1分数较低，这与神经网络在标签稀疏数据上的已知表现一致。 p值对应均值的Wilcoxon符号秩检验，显著性水平为0.05。

我们对两个选定的感兴趣检测方法的模型表现进行了进一步评估：PXR 激动作用（在 Tox21 中标记为 `tox21-pxr-p1` ）和 HepG2 细胞活力（`tox21-rt-viability-hepg2-p1`）。我们选择这些检测方法有两个原因：（1）它们在语义上与所有其他 Tox21 检测方法明显不同（即没有其他检测方法测量孕烷 X 活性或细胞活力），因此我们不期望信息泄露。

As shown in **Figure 3**, the GNN model significantly outperforms both the random forest (Wilcoxon signed-rank test p-value $2.3 \cdot 10^{-4}$) and gradient boosting (p-value $2.6 \cdot 10^{-3}$) models in terms of area under the receiver operating characteristic curve (AUROC), with a mean AUROC of 0.883 (compared to 0.834 for random forest and 0.851 for gradient boosting). This is robust evidence that the GNN model tends to substantially outperform 'traditional' QSAR models. A notable characteristic of the GNN AUROCs is that their distribution has a higher variance than either the random forest or gradient boosting AUROCs. Anecdotally, this is likely due to diminished sensitivity of the GNN model when trained on assays with few positive examples—neural networks tend to struggle as data become more sparse, which seems to be the case here. We also compared F1-score distributions between the 3 model types; however, the differences between the 3 models are not statistically significant. The relatively low F1-scores in the 3 model types is a result of the class imbalance in the QSAR toxicity assays—all of the assays contain far more negative samples (assay is inactive) than positive samples (assay is active), which results in any false negatives having a magnified impact on F1. The same increased variance observed in GNN model AUROCs is shown in the GNN F1-scores.

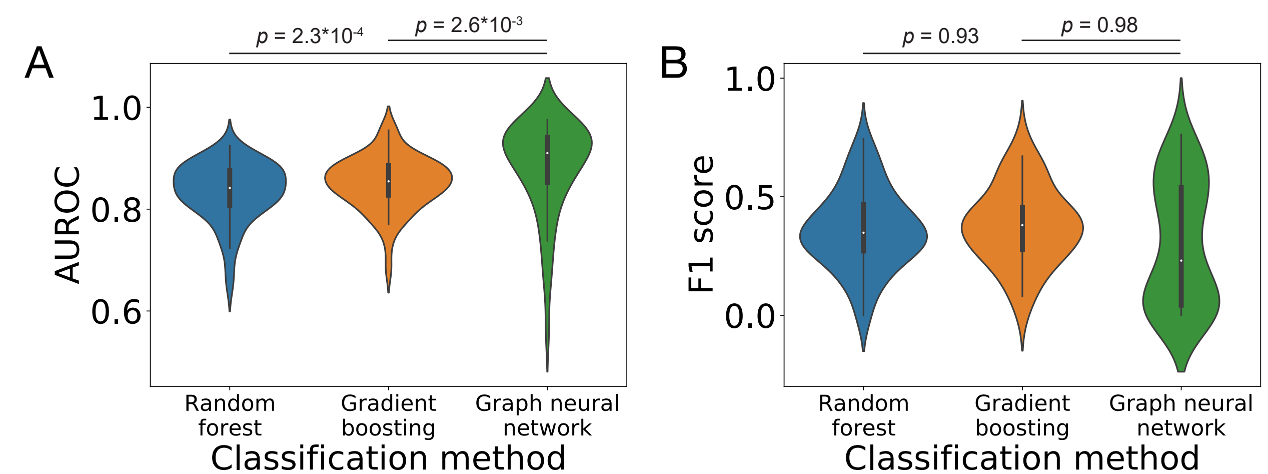

Fig. 3. Overall performance metrics of the 3 QSAR model types on each of the Tox21 assays—a.) area under the receiver operating characteristic curve (AUROC) and b.) F1 score. The mean AUROC is significantly higher for the GNN model than for either of the two baseline QSAR approaches. The differences in F1 scores are not statistically significant. The GNN achieves poor F1 scores on assays with relatively few (e.g., < 100) "active" annotations in Tox21, which is consistent with known performance of neural networks on data with sparse labels. p-values correspond to Wilcoxon signed-rank tests on means, with a significance level of 0.05.

We performed further review of model performance on two selected assays of interest: PXR agonism (labeled `tox21-pxr-p1` in Tox21) and HepG2 cell viability (`tox21-rt-viability-hepg2-p1`). We selected these assays for two reasons: (1.) Both are semantically distinct from all other Tox21 assays (i.e., there are no other assays measuring pregnane X activity or cell viability), and therefore we would not expect an information leak

来自GNN中其他高度相关的Tox21实验，以及(2)两者都有足够数量的阳性化学物质，以便它们的ROC曲线在三种模型类型的所有决策规则值上获得高分辨率。图4显示，在几乎所有判别阈值下，GNN在两种情况下都优于随机森林和梯度提升模型。GNN在HepG2细胞活力上的高性能尤其值得注意——在化学筛选实验中，细胞活力历来难以预测。其他50个Tox21实验中的许多也显示出类似的表现模式。所有ROC图均可在补充资料中获取。

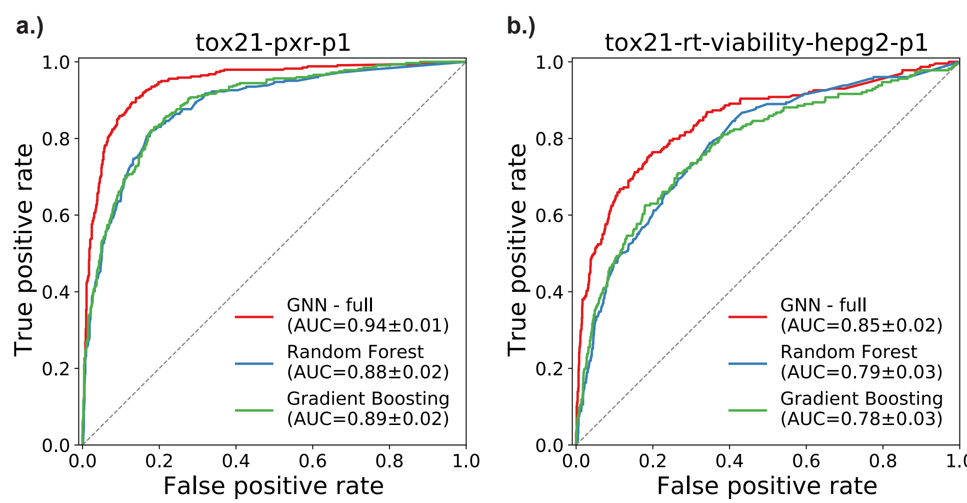

图4. 两个选定Tox21测定的受试者工作特征（ROC）曲线：a.) PXR激动（tox21-pxr-p1）和 b.) HepG2细胞活力（tox21-rt-viability-hepg2-p1）。在这两种情况下，GNN模型的曲线下面积（AUC）显著高于随机森林或梯度提升模型。AUC值以95%的置信区间给出。

3.2. 图组件对已训练预测模型影响的消融分析

为了更好地理解GNN模型是如何优于随机森林和梯度提升模型的，我们对前面提到的两种测定进行了消融分析——孕烷X激动作用和HepG2细胞活力。对于这两种测定，我们在移除GNN的特定组件后重新训练了模型：

- 所有检测节点。
- 所有基因节点。
- 化学节点的MACCS指纹（用虚拟变量替换它们，以便网络的结构保持不变）。

这些实验的ROC曲线如图5所示。对于这两种测定，完整的GNN模型表现最好，尽管仅在AUROC方面略优于没有MACCS指纹或基因节点的版本。然而，GNN的性能有所下降

from other highly correlated Tox21 assays present in the GNN, and (2.) both have a sufficient number of positive chemicals such that their ROC curves attain high resolution at all values of the decision rule across the 3 model types. **Figure 4** shows that the GNN outperforms the random forest and gradient boosting models at virtually all discrimination thresholds in both cases. The high performance of the GNN on HepG2 cell viability is especially noteworthy—cell viability is notoriously challenging to predict in chemical screening experiments. Many of the other 50 Tox21 assays showed similar patterns in performance. All ROC plots are available in **Supplemental Materials**.

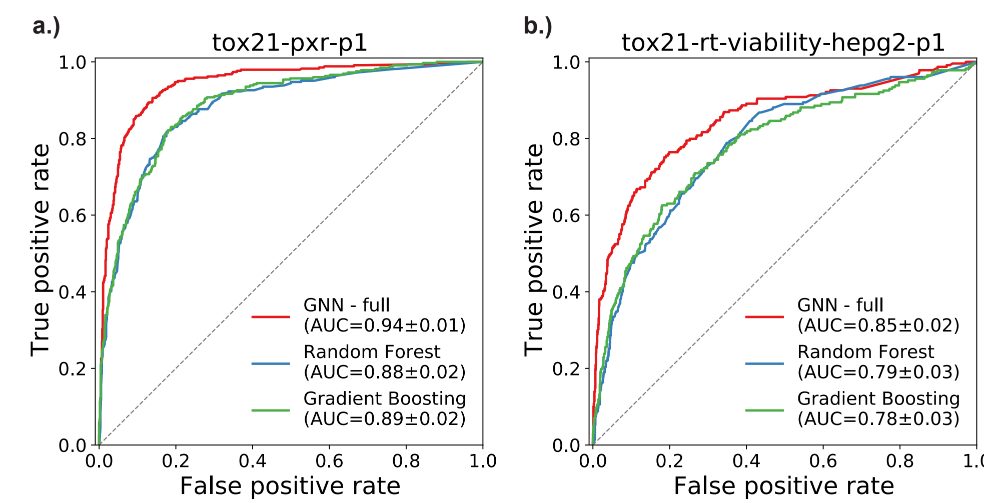

Fig. 4. Receiver operating characteristic (ROC) curves for two selected Tox21 assays: a.) PXR agonism (`tox21-pxr-p1`) and b.) HepG2 cell viability (`tox21-rt-viability-hepg2-p1`). In both cases, the area under the curve (AUC) is significantly higher for the GNN model than either the Random Forest or Gradient Boosting models. AUC values are given with 95% confidence intervals.

3.2. *Ablation analysis of graph components' influence on the trained predictive model*

To better understand how the GNN model outperforms the random forest and gradient boosting models, we performed an ablation analysis on the two previously mentioned assays—pregnane X agonism and HepG2 cell viability. For both of the assays, we re-trained the model after removing specific components from the GNN:

- All assay nodes.
- All gene nodes.
- MACCS fingerprints for chemical nodes (replacing them with dummy variables so the structure of the network would remain the same).

ROC plots for these experiments are shown in **Figure 5**. For both assays, the full GNN model performed best, although only modestly better (in terms of AUROC) than the versions without MACCS fingerprints or gene nodes. However, the performance of the GNN drops

实质上——几乎仅比随机猜测标签稍好（随机猜测标签对应的AUROC为0.50）——当测定节点从图中移除时。换句话说，GNN模型的大部分推理能力来自化学物质与除正在预测活性之外的其他测定之间的连接。同样，MACCS指纹本身并不足以让GNN达到与仅使用MACCS指纹作为预测特征的基线QSAR模型相同的性能。因此，尽管GNN比两个基线模型的性能显著更好，但它只有在加入化学物质、测定和（在较小程度上）基因之间的网络关系的上下文时，才能够实现这一点。

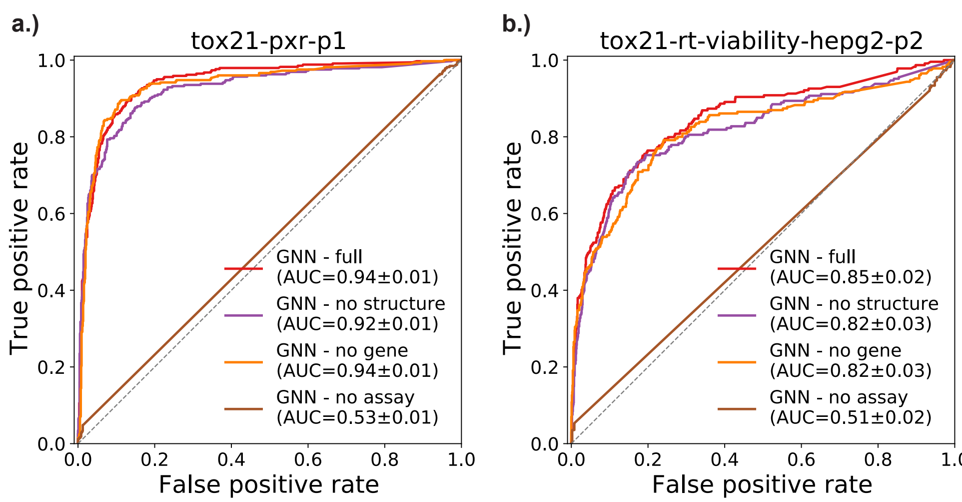

图5. 使用GNN模型不同配置对两个选定的Tox21测定进行的受试者工作特征（ROC）曲线。“GNN - full”是§2.3.1中描述的完整模型。“GNN - no structure”省略了MACCS化学描述符，并用相同维度的节点嵌入替代。“GNN - no gene”移除了基因节点及其相关边。“GNN - no assay”移除了所有测定节点及相关边，因此预测仅使用化学物质、基因、剩余边以及作为化学节点特征的MACCS化学描述符。AUC值给出了95%的置信区间。

4. 讨论

4.1. 用于***QSAR***建模的***GNN***与传统机器学习

毒理学界普遍认为，QSAR 在许多任务上表现不佳，迫切需要新的方法学进展。在本研究中，我们展示了 GNN 在该领域显著优于当前的金标准技术。除了神经网络比非神经网络模型更容易适应非线性目标的事实之外，这很可能是纳入超越化学结构特征的生物医学知识的自然结果。基因相互作用提供了有关化学物质如何影响体内代谢和信号通路的线索，而非靶点检测（即图中除当前预测的检测之外的其他检测）可能与

substantially—barely better than guessing labels at random (which would correspond to an AUROC of 0.50)—when assay nodes are removed from the graph. In other words, much of the inferential capacity of the GNN models are conferred by chemicals' connections to assays other than the one for which activity is being predicted. Similarly, MACCS fingerprints are not—on their own—enough for the GNN to attain equal performance to the baseline QSAR models, which only use MACCS fingerprints as predictive features. Therefore, although the GNN achieves significantly better performance than the two baseline models, it is only able to do so with the added context of network relationships between chemicals, assays, and (to a lesser degree) genes.

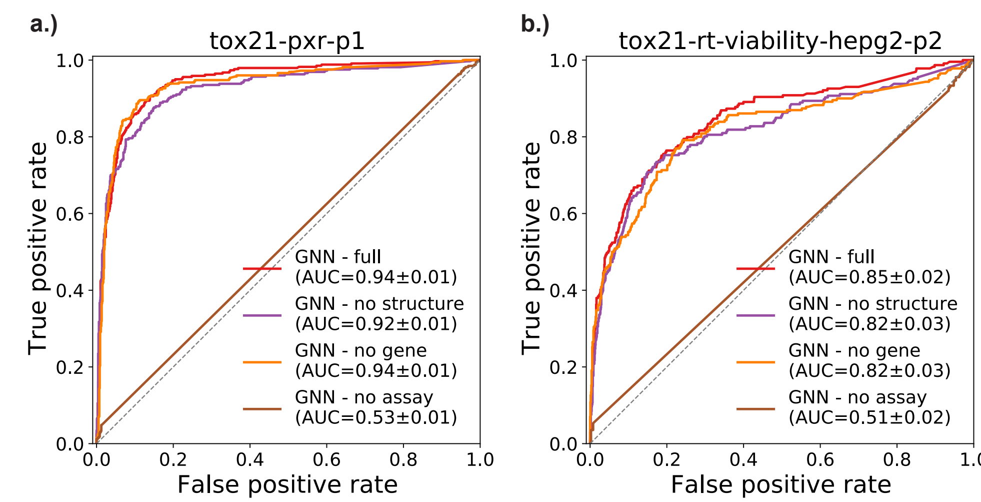

Fig. 5. Receiver Operator Characteristic (ROC) curves for two selected Tox21 assays using different configurations of the GNN model. 'GNN - full' is the complete model as described in §2.3.1. 'GNN - no structure' omits the MACCS chemical descriptors and replaces them with node embeddings of the same dimensionality. 'GNN - no gene' removes gene nodes and their incident edges from the network. 'GNN - no assay' removes all assay nodes and incident edges, so predictions are made solely using chemicals, genes, the remaining edges, and the MACCS chemical descriptors as chemical node features. AUC values are given with 95% confidence intervals.

4. Discussion

4.1. *GNNs versus traditional ML for QSAR modeling*

The toxicology community largely agrees that QSAR underperforms on many tasks, and that new methodological advances are desperately needed. In this study, we demonstrate that GNNs significantly outperform the current gold-standard techniques in the field. Aside from the fact that neural networks can more easily adapt to nonlinear objectives than non-neural network models,[20] this is likely a natural consequence of incorporating biomedical knowledge that goes beyond chemical structure characteristics. Gene interactions provide clues about how chemicals influence metabolic and signaling pathways *in vivo*, and non-target assays (i.e., other assays in the graph aside from the one currently being predicted) may correlate with

目标测定的活性。

4.2. *QSAR* 中 *GNN* 的可解释性

化学指纹——例如我们在本研究中使用的MACCS——提供了一种适合机器学习的化学物质表示方法。然而，基于指纹的模型难以解释。虽然MACCS指纹的每个字段对应于有意义的化学属性（例如化学物质是否含有多个芳香环，或者至少一个氮原子），但在QSAR应用中，指纹在很大程度上是难以理解的，因为生物活性是目标化学物质与生物分子之间许多高阶相互作用的结果。

在这项研究中，基于知识表示的异构图数据可以轻松表示在化学物质导致的毒性反应中起中介作用的实体类型之间的可解释关系。虽然在这项特定研究中并未实现，但一种被称为图注意网络（Graph Attention Network, GAT）的GNN架构可以明确突出图中在预测中具有影响力的部分，这为继续在QSAR建模中研究GNN提供了逻辑上的下一步。其他更简单的方法也为探索可解释性提供了途径，例如可视化训练好的GNN中从检测节点出发的边的权重。通常，图的巨大的规模使这种方法难以实现，但由于我们的图仅包含52个检测，因此检查它们的权重相对简单。例如，对于HepG2细胞活力预测任务，权重最高的检测是HepG2 Caspase-3/7介导的细胞毒性以及NIH/3T3 Sonic hedgehog拮抗（发育毒性标志）。

我们在补充材料中提供了上述两种测定的所有分析重量。

4.3. 所以偏差来源及其对毒性预测*QSAR*的影响

像任何元分析技术一样，QSAR 也容易受到多种偏差的影响，这些偏差可以在多个层面上引入，其中尤以用于生成训练数据样本的毒性活性注释的原始实验为甚。历史上这是一个更大的问题，当时化学物质的已知活性要么汇编自已发表的科学期刊文章结果，要么来自体内实验的报告指南。出版偏差导致负活性注释极不完整，而用于推断负面注释的技术也不一致。较早的 QSAR 研究通常未说明其数据的原始来源，因此结果的验证和可重复性非常具有挑战性（如果不是不可能的话）。

幸运的是，现代大规模筛选工作（包括Tox21）是为了直接解决这些和其他问题而创建的。[24] 虽然我们的训练数据仍然受到批次效应、在选择用于筛选的试验和化学品时的偏差以及其他系统性和实验性错误的影响

activity of the target assay.

4.2. *Interpretability of GNNs in QSAR*

Chemical fingerprints—such as MACCS, which we use in this study—provide a valuable approach to representing chemicals that is suitable for machine learning. However, models based on fingerprints are challenging to interpret.[7,21] Although each field of a MACCS fingerprint corresponds to meaningful chemical properties (such as whether the chemical contains multiple aromatic rings, or at least one nitrogen atom), the fingerprint is largely inscrutable in QSAR applications, since biological activity is the result of many higher-order interactions between the chemical of interest and biomolecules.

In this study, the knowledge representation-based heterogeneous graph data represent easily interpretable relationships between entity types that mediate toxic responses to chemicals. Although not implemented in this particular study, a GNN architecture known as a *graph attention network* explicitly highlights portions of a graph that are influential in predictions, providing a logical next step for continuing this body of work on GNNs in QSAR modeling. Other, simpler approaches also provide avenues for exploring interpretability, such as visualizing the edge weights for edges starting at assay nodes in the trained GNN. Often, the sheer size of graphs make this approach intractable, but since our graph only contains 52 assays it is relatively straightforward to inspect their weights. For example, the highest weighted assays for the HepG2 cell viability prediction task are *HepG2 Caspase-3/7 mediated cytotoxicity* and *NIH/3T3 Sonic hedgehog antagonism* (a marker of developmental toxicity). The first of these makes sense from an intuitive standpoint, as it measures toxic response in the same cell line as the predicted assay. The second, on the other hand, does not have an immediately obvious connection to the predicted assay, but may be linked to the fact that Shh antagonists can induce apoptosis.[22] Either way, it is easy to see that assay weights can be used to generate specific hypotheses for future targeted studies of mechanisms that underlie toxicity.

We provide all assay weights for the two above-mentioned assays in **Supplemental Materials**.

4.3. *Sources of bias and their effects on QSAR for toxicity prediction*

Like any meta-analysis technique, QSAR is subject to multiple sources of bias that can be introduced at several levels, not the least of which is in the original experiments used to generate toxic activity annotations for training data samples. This was a greater issue historically, when known activities for chemicals were compiled either from published scientific journal article results or from reporting guidelines for *in vivo* experiments.[23] Publication bias caused negative activity annotations to be extremely incomplete, and techniques for imputing negative annotations were inconsistent. Older QSAR studies often did not state the original sources of their data, so verification and reproducibility of results are immensely challenging (if not impossible).

Fortunately, modern large-scale screening efforts (including Tox21) were created to directly address these and other issues.[24] While our training data are still subject to batch effects, bias in selecting assays and chemicals for screening, and other systematic and experimental errors

这些被传播到最终QSAR模型中的因素，让我们相对有信心，早期QSAR研究中困扰的问题，如出版偏倚、报告偏倚等，已经大幅减少。此外，我们的GNN方法用于QSAR建模，可能比非GNN方法对这些偏倚来源更具鲁棒性，因为（a）图形整合了多层次的生物学知识，可以“填补”其他层次中不完整或不准确数据留下的空白；（b）GNN——尤其是异质GNN——表现出一些性质，使其本质上对噪声具有鲁棒性。

5. 结论

在本研究中，我们介绍了一种用于毒性预测的基于GNN的QSAR建模新方法，并在52个实验的数据上进行了评估，结果显示其性能显著优于现有方法。GNN是人工智能研究中一个非常活跃的新兴主题，作为计算毒理学中最早的GNN应用之一，我们希望我们的结果能够作为类似工作的“出发点”。我们计划在不久的将来评估图注意力网络、新的数据模态和网络正则化技术，并鼓励广泛的毒理学和信息学社区为改善预测毒理学的整体数据生态系统贡献力量。

6. 代码可用性

与本研究相关的所有源代码可在 GitHub 上获得，网址为 `https://github.com/EpistasisLab/qsar-gnn`。在撰写本文时的代码冻结副本可在 `https://doi.org/10.5281/zenodo.5154055` 获得。

7. 补充材料

补充表格和图表可在 FigShare 上获取，网址为 `https://doi.org/10.6084/m9.figshare.15094083`。

致谢

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附录A. 图卷积网络架构

我们的 GCNN 实现使用了一种消息传递范式，结合了 GraphSAGE[27] 和 R-GCN[16] 架构的特性。设 $G = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ 为一个异质图，由节点 $v_i \in \mathcal{V}$、边 $(v_i, r, v_j) \in \mathcal{E}$ 以及一组边类型 $r \in \mathcal{R}$ 组成。每条边都标注着恰好一种边类型。所有化学节点（下文用 h^0 表示）由长度为 166 的二进制字符串表示，对应化学物质的 MACCS 指纹，而所有其他节点（测定和基因）由在优化过程中学习的单个十进制值“嵌入特征”表示。测定或基因嵌入的大小为

that are propagated along to the final QSAR model, we are relatively confident that publication bias, reporting bias, and other issues that plagued early QSAR studies have been substantially decreased. Furthermore, our GNN approach to QSAR modeling may be more robust to these sources of bias than non-GNN approaches, because (a.) the graph incorporates multiple levels of biological knowledge that can ‘fill in gaps’ left by incomplete or inaccurate data at other levels and (b.) GNNs—and heterogeneous GNNs in particular—exhibit properties that make them inherently robust to noise.[25,26]

5. Conclusions

In this study, we introduce a novel GNN-based approach to QSAR modeling for toxicity prediction, and evaluate it on data from 52 assays to show that it significantly outperforms existing methods. GNNs comprise an incredibly active emerging topic within artificial intelligence research, and as one of the first GNN applications in computational toxicology we hope that our results serve as a ‘jumping off point’ for a vast body of similar work. We plan to evaluate graph attention networks, new data modalities, and network regulization techniques in the near future, and encourage contributions from the toxicology and informatics communities at-large to improve predictive toxicology’s overall data ecosystem.

6. Code availability

All source code pertaining to this study is available on GitHub at `https://github.com/EpistasisLab/qsar-gnn`. A frozen copy of the code at the time of writing is available at `https://doi.org/10.5281/zenodo.5154055`.

7. Supplemental Materials

Supplemental tables and figures are available on FigShare at `https://doi.org/10.6084/m9.figshare.15094083`.

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Appendix A. Graph convolutional network architecture

Our GCNN implementation uses a message-passing paradigm that combines aspects of the GraphSAGE[27] and R-GCN[16] architectures. Let $G = (\mathcal{V}, \mathcal{E}, \mathcal{R})$ be a heterogeneous graph consisting of nodes $v_i \in \mathcal{V}$, edges $(v_i, r, v_j) \in \mathcal{E}$, and a set of *edge types* $r \in \mathcal{R}$. Each edge is labeled with exactly one edge type. All chemical nodes (represented below as h^0) are represented by a bit string of length 166 corresponding to the chemical’s MACCS fingerprint, while all other nodes (assays and genes) are represented by a single decimal-valued ‘embedding feature’ that is learned during optimization. The magnitude of an assay or gene’s embedding is

大致相当于该节点在网络中的重要性，并且可以用于模型解释的内省。

网络的每一层被定义为相邻节点的边缘聚合：

$$h_i^{(l)} = \sigma\left(\sum_{r\in\mathcal{R}} \rho_{j\in\mathcal{N}_i^r}\left(W_r^{(l-1)}h_j^{(l-1)} + W_0^{(l-1)}h_i^{(l-1)}\right)\right). \tag{A.1}$$

其中 h_i^l 是第 l 层节点 i 的隐藏表示，$\mathcal{N}(i)$ 是节点 i 的直接邻居集合，σ 是非线性激活函数（如附录 B 所述，可以是 softmax 或 leaky ReLU）。ρ 可以是任何将来自单一类型的连边传递的信息进行组合的可微分“归约”函数；在本研究中，我们使用求和。由于我们的图包含的边类型相对较少，因此不需要对权重矩阵 W 进行正则化。

附录 B. 节点分类模型

为了将化学物质根据感兴趣的测定分为活性或非活性，我们按（A.1）所示的形式堆叠两个 GCN 层，并在两层之间使用泄漏 ReLU 激活函数，同时对第二层的输出应用 softmax。由于我们只对化学节点进行分类，因此忽略所有其他节点类型（以及标记未定义的化学物质）的输出；标签通过算法 1 生成。然后我们通过最小化网络 softmax 输出与真实活性值之间的二元交叉熵来训练网络：

$$\mathcal{L} = -\sum_{i\in\mathcal{Y}} \ell(h_i^{(0)})\cdot\ln h_i^{(2)} + (1-\ell(h_i^{(0)}))\cdot\ln(1-h_i^{(2)}). \tag{B.1}$$

其中 $\mathcal{Y}$ 是所有有标签节点的集合，$\ell(h_i^{(0)})$ 是节点 i 的真实标签，$h_i^{(2)}$ 是节点 i 的最终层输出。

网络相对浅显的架构使我们能够使用 Adam 算法对整个训练数据集进行模型优化，但在适当或必要时，模型也可以调整为小批量训练。

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roughly equivalent that node's importance in the network, and can be introspected for model interpretation.

Each layer of the network is defined as an edge-wise aggregation of adjacent nodes:

$$h_i^{(l)} = \sigma\left(\sum_{r\in\mathcal{R}} \rho_{j\in\mathcal{N}_i^r}\left(W_r^{(l-1)}h_j^{(l-1)} + W_0^{(l-1)}h_i^{(l-1)}\right)\right). \tag{A.1}$$

where h_i^l is the hidden representation of node i in layer l, $\mathcal{N}(i)$ is the set of immediate neighbors of node i, and σ is a nonlinear activation function (either softmax or leaky ReLU, as explained in **Appendix B**). ρ can be any differential 'reducer' function that combines messages passed from incident edges of a single type; in the case of this study we use summation. Since our graph contains relatively few edge types, regularization of the weight matrices W is not needed.

Appendix B. Node classification model

For classifying chemicals as active or inactive with regards to an assay of interest, we stack 2 GCN layers in the form given by (A.1), with a leaky ReLU activation between the two layers and softmax applied to the second layer's output. Since we only classify chemical nodes, we ignore outputs for all other node types (and for chemicals with undefined labels); labels are generated via **Algorithm 1** We train the network by minimizing binary cross-entropy between the network's softmax outputs and true activity values:

$$\mathcal{L} = -\sum_{i\in\mathcal{Y}} \ell(h_i^{(0)})\cdot\ln h_i^{(2)} + (1-\ell(h_i^{(0)}))\cdot\ln(1-h_i^{(2)}). \tag{B.1}$$

where $\mathcal{Y}$ is the set of all labeled nodes, $\ell(h_i^{(0)})$ is the true label of node i, and $h_i^{(2)}$ is the final layer output of node i.

The relatively shallow architecture of the network allows us to optimize the model using the Adam algorithm applied to the entire training data set, but the model can be adapted to mini-batch training when appropriate or necessary.

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